CS 532, Spring 2019: Quiz 6

NAME:

Problem 1. Consider a binary image (black and white) where only a triplet of intersecting black lines, and a single isolated point are visible. One line is vertical ℓ_v, one line is horizontal ℓ_h, while the other line ℓ_s has a 45 degree angle; all lines intersect in a single vertex point $\mathbf{v}$. The individual point $\mathbf{p} = (x, y, 1)$ is a single black pixel surrounded by white pixels. Answer the following:

- Write the expression relating the coordinates of vertex of intersection w.r.t. the horizontal and vertical lines.
- Would the single point of intersection be a good feature to track? Explain why.
- Consider an arbitrary point along the slanted line, having strong gradients along both image axis directions. Would such a point be a good feature to track? Explain why.
- Would the location of the single individual point be any better feature to track than the location of a white pixel right next it? Explain why.